

SHETH L.U.J AND SIR M.V COLLEGE

SUBJECT: Artificial intelligence
PRACTICAL NO. 4

Aim: Implement the Feed Forward Backpropagation algorithm to train a neural network.

- Use a given dataset to train the neural network for a specific task.
- Evaluate the performance of the trained network on test data.

CODE:

```
import matplotlib.pyplot as plt
import pandas as pd

from sklearn.metrics import accuracy_score, classification_report
from sklearn.model_selection import train_test_split
from sklearn.neural_network import MLPClassifier
from sklearn.preprocessing import StandardScaler

# Load Iris dataset
df = pd.read_csv("Iris.csv")

# Remove extra spaces from column names
df.columns = df.columns.str.strip()

# Separate input features and target
X = df.drop(["Species", "Id"], axis=1)
y = df["Species"]

# Split the dataset into training and testing data
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42,
    stratify=y
)

# Standardize the input features
scaler = StandardScaler()

X_train = scaler.fit_transform(X_train)
```

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```
X_test = scaler.transform(X_test)

# Feed Forward Backpropagation Neural Network
model = MLPClassifier(
    hidden_layer_sizes=(6, 8),
    activation="relu",
    solver="adam",
    max_iter=300,
    random_state=42
)

# Train the neural network
model.fit(X_train, y_train)

# Predict test data
y_pred = model.predict(X_test)

# Evaluate the model
print("Accuracy:", accuracy_score(y_test, y_pred) * 100, "%")

print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Plot the loss curve
plt.plot(model.loss_curve_)

plt.title("Feed Forward Backpropagation Loss Curve")
plt.xlabel("Iteration")
plt.ylabel("Loss")

plt.show()
```

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OUTPUT:

